Deliverable 2

Course Instructor

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Submitted By

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UC01	
Use case:	Manage Employee Accounts
Actors:	Admin
Type:	Primary
Description:	The admin logs in to the system and chooses the option to manage the employee. The admin can perform actions such as adding, editing, or deleting the employee's account. The system processes the admin's request and updates the employee's information accordingly. Upon success, it displays a success message.

UC02	
Use case:	Manage Inventory
Actors:	Admin, Employee
Type:	Primary
Description:	The user will log in to the system and access the "inventory management" module. The user can add a new item to the inventory, edit an existing item in the inventory, or delete an existing item in the inventory. The system will perform the respective action. The system will verify the information provided by the user about the item. After performing the operation, it will display a message indicating that the operation was successful. Then it will update the record in the inventory.

UC03	
Use case:	Receive Stock Alerts
Actors:	Admin
Type:	Primary
Description:	The admin logs into the system. Then the admin goes to the inventory management section. The system will retrieve all the information regarding the inventory from the database. If the stock is out, it will send an alert. Then the alert will be sent to the admin after the task is done, and the alert will be removed.

UC04	
Use case:	Filter Sales Data
Actors:	Admin
Type:	Primary
Description:	The admin logs in, selects the option "Filter Sales Data" under the sales module, and filters the sales data. The filter analysis allows the admin to analyze specific data, like dates, product categories, or store locations, from the sales data. The system processes the filter criteria, and the filtered data displays the specific data to be analyzed.

UC05	
Use case:	Set Monthly Sales Target
Actors:	Admin
Type:	Primary
Description:	The admin will log in to the system and select the Set Monthly Sales Targets option. The admin will enter the values for employee targets. The system will validate the values for employee targets for future performance tracking.

UC06	
Use case:	Review Employee Performance
Actors:	Admin
Type:	Primary
Description:	The admin logs in to the system, and they click "Review Employee Performance." The admin then selects an employee to review, and the system fetches the data to show the employee's performance metrics and comparisons.

UC07	
Use case:	View Sales Dashboard
Actors:	Admin
Type:	Primary
Description:	The admin logs into the system and selects the Sales Dashboard option. The system retrieves analytics data and displays visual summaries such as charts, statistics, and key performance indicators showing overall sales performance.

UC08	
Use case:	Generate Monthly Reports
Actors:	Admin
Type:	Primary
Description:	The admin logs into the system and selects the Generate Monthly Reports option. The admin chooses the desired month and report type. The system compiles the relevant sales data and generates a report that can be viewed or exported.

UC09	
Use case:	Process Sale
Actors:	Employee
Type:	Primary
Description:	The user creates a new billing session for a customer at a point of sale. The user then proceeds to add items to the bill using a barcode scanner or by searching for the items. This system will then keep a record of all items added to the bill, calculate the running total, as well as calculate any taxes and discounts. The user will have the option to add quantities of items and remove items before checking out.

UC10	
Use case:	Process Item Return
Actors:	Employee
Type:	Primary
Description:	The Employee initiates the return of the items on behalf of the customer who returns a previously purchased products. The Employee retrieves the original transaction from the bill history and picks the items to be returned. Then, the system checks the eligibility for the return and proceeds to return the items. After a successful return, the system adds the returned item back to the inventory and records a return transaction.

UC11	
Use case:	Generate Customer Bill
Actors:	Employee
Type:	Primary
Description:	After all the items are included in the 'Customer Bill Session', the Employee requests the system to generate the final bill for the customer. The system calculates the total amount including taxes and generates a receipt in a formatted manner. The Employee presents the bill to the customer in the form of a receipt or in digital form. After the bill is generated, it is stored in the history.

UC12	
Use case:	View Inventory Availability
Actors:	Admin, Employee
Type:	Primary
Description:	The user accesses the inventory module to view the existing stock levels of the products. The system provides the user with real-time data of the existing inventories, i.e., the name of the products, categories of the products, the quantity of the products available, and the status of the products. Upon completing the action, the actor is notified of the existing stock levels of the products.

UC13	
Use case:	Access Bill History
Actors:	Admin, Employee
Type:	Primary
Description:	The user uses the bill history module to access the bill history records. The system will display a list of completed bills that can be filtered by date, employee, or amount. The user can choose a bill to view all the information related to the bill. Once done, the actor has reviewed the necessary transaction records for return, audit, or sales analysis.

UC14	
Use case:	Track Personal Performance
Actors:	Employee
Type:	Primary
Description:	The employee logs in to the system and selects the option to view personal performance. The system retrieves the employee's sales records and calculates performance indicators such as total sales and progress toward monthly targets. On success, the employee is shown a summary report of their performance.

UC15	
Use case:	Manage Product Categories
Actors:	Admin, Employee
Type:	Primary
Description:	The user logs in the system and selects the product category management option in the system. The system displays the current list of product categories. The user performs a CRUD operation, and the system validates the requested changes and updates the category records accordingly. The updated list of product categories is then displayed to the actor. On Success, the user is shown a success message, and the categories are updated in the database.

UC01

Use Case Section	Comment
Use Case Name*	Manage Employee Accounts
Scope	SmartKiryana
Level	User Goal
Primary Actor*	Admin
Stakeholders and Interests	• **Admin:** Wants to Apply CRUD Operation to Employee for contained access and up-to-date information • **Employee:** Want to get registered or change current info
Preconditions	• Admin Should be Logged in the system
Success Guarantee (Post Conditions)	• The records are updated in the database • A success message is displayed
Is Main Success Scenario*	**UC01: Main Success Scenario**

Actor Action	System Responsibility
1. Admins logs in the system and chooses the Manage Employees Option	
	2. The System Shows the current Employee List
3. Admin chooses the desired CRUD operation and inputs any information required.	3. The system Validates the information entered
	4. The System updates the employee Database.
	5. A Success Message is shown to the admin

Extensions*	• The Employee details are incomplete
	  ○ Ask the user to enter complete Details
	• The Employee already exists
	  ○ Ask the user to enter new Employee details
	• Database not connected
	  ○ Reload the Manage Employee Module
	  ○ Contact IT Department

Special Requirements	• The Data entered should be stored securely and only accessible to admin • System must respond to operations within 3 seconds.
Technology and Data Variations List	• The Employee may provide his CNIC Number or Passport for identification
Frequency of Occurrence	• Occasionally. Used when Data of Employee Change
Miscellaneous	• Are there any roles between the Employees? • Could we integrate **employee attendance modules** in the future? • Could the system include **employee account deactivation** instead of permanent deletion?

UC02

Use Case Section	Comment
Use Case Name*	Manage Inventory
Scope	SmartKiryana
Level	User Goal
Primary Actor*	Admin, Employee
Stakeholders and Interests	**Admin:** Wants the Inventory to be up-to date **Employees**: Wants the Inventory to be up-to date to process sale smoothly
Preconditions	• The user must be successfully logged into the system.
Success Guarantee (Post Conditions)	• Inventory records are updated in the database according to the performed operation. • The system displays a Success message
Main Success Scenario*	**UC02: Main Success Scenario**

Actor Action	System Responsibility
1. The user logs into the system and navigates to the Inventory Management module.	
	2. The system displays the list of inventory items along with available operations.
3. The user selects one of the Available operations and enters the required information.	3. The system validates the entered information
	4. The system updates the

	inventory records in the database according to the selected operation.
	5. The system displays a success message confirming the operation.

Extensions*	• Invalid or Incomplete Data
	  ○ Ask the user to enter correct data.
	• Duplicate Item Entry
	  ○ Notify the User the Item already Exists
	• Database or System Error
	  ○ Ask the User to try again
	  ○ On persistent Issue contact the IT department

Special Requirements	• Only authorized users should be allowed to use the inventory management operations. • The interface will show the user the inventory information. It will also allow easy navigation for the operations. • The system will ensure that the inventory is updated atomically.
Technology and Data Variations List	• Item identifier entered by bar code laser scanner (if bar code is present) or keyboard.
Frequency of Occurrence	• Daily. Used Almost daily whenever new batch of items arrive
Miscellaneous	• Explore the **bulk inventory import/export** (e.g., CSV or Excel files) option • Should employees have **limited permissions compared to admins** (e.g., cannot delete items)?

UC03

Use Case Section	Comment
Use Case Name*	Receive Stock Alerts
Scope	SmartKiryana.
Level	User Goal
Primary Actor*	Admin
Stakeholders and Interests	• **Admin:** Wants to be quickly informed about out-of-stock items so that necessary actions (such as restocking) can be taken.
Preconditions	• The admin must be successfully logged into the system. • Inventory records must exist in the system.
Success Guarantee (Post Conditions)	• The admin is notified about all out-of-stock items. • Alerts are cleared for the current login session after being displayed.
Main Success Scenario*	**UC03: Main Success Scenario**

Actor Action	System Responsibility
1. The admin logs into the system and proceeds to the Inventory Management module.	
	2. The system retrieves inventory records from the database and analyses inventory quantities for each item.
	3. The system identifies items with few stock quantities and displays alerts for these items.

	4. The admin acknowledges the alerts.	4. The system clears up the alerts for the current login session.

Extensions*	• No Out-of-Stock Items
	  ○ The system shows no alert
	• System Failure
	  ○ The user restarts the system and logs in again
	  ○ If the issue persists, contact the IT department	

Special Requirements	• The system should display stock alerts within 2 seconds when accessing the inventory module. • The stock alerts should contain accurate information from the database concerning the stock. • The stock alerts should be visible.	
Technology and Data Variations List	• Alerts may be displayed as pop-up notifications, dashboard messages, or highlighted inventory items.	
Frequency of Occurrence	• Medium – Occurs only when a registered Item is low on Inventory.	
Miscellaneous	• Should the system support **automatic email or SMS notifications** for out-of-stock items? • Should there be configurable stock thresholds?	

UC04

Use Case Section	Comment
Use Case Name*	Filter Sales Data
Scope	SmartKiryana
Level	User Goal
Primary Actor*	Admin
Stakeholders and Interests	**Admin:** Wants quick filtered insights for decision making. **Sales Analyst:** Needs precise data for analysis.
Preconditions	• Actor is logged into the system. • Sales data exists in the database.
Success Guarantee (Post Conditions)	• The system displays filtered sales records based on the criteria.
Main Success Scenario*	**UC04: Main Success Scenario**

Actor Action	System Responsibility
1. Admin opens the sales data module.	1. System displays all available sales data.
2. Admin selects filter criteria (date range, product category, store location).	
	3. System receives filter parameters.
4. Admin applies the filter.	4. System processes the filtering request.
	5. System displays the filtered sales data.

Extensions*	Invalid filter criteria entered • System displays error message and requests correction. No matching data found • System displays message: No records found
Special Requirements	• Filtering results should be displayed within 2 seconds for good user experience. • Data must be presented in sortable tables or charts
Technology and Data Variations List	• Filters may include date ranges, product categories, store locations, or employee IDs. • Data can be displayed in table view or graphical analytics view.
Frequency of Occurrence	• High – Actors may perform this operation multiple times daily.
Miscellaneous	• Future versions may allow saving custom filter presets.

UC05

Use Case Section	Comment
Use Case Name*	Set Monthly Sales Targets
Scope	SmartKiryana
Level	User Goal
Primary Actor*	Admin
Stakeholders and Interests	**Admin:** Needs to define achievable sales goals. **Employees:** Require clear performance targets.
Preconditions	• Admin is authenticated in the system.
Success Guarantee (Post Conditions)	• Monthly sales targets are stored successfully.
Main Success Scenario*	**UC05: Main Success Scenario**

Actor Action	System Responsibility
1. Admin opens the sales target module.	
	2. System displays target configuration interface.
3. Admin selects employee.	3. System retrieves relevant employee information.
4. Admin enters monthly sales targets.	4. System validates entered values.
5. Admin confirms the targets.	5. System saves targets in the database.
	6. System displays success message.

Extensions*	Invalid target value entered • System prompts admin to correct the value.
Special Requirements	• Target values must be numerical and within predefined limits. • Only authorized actors can set or modify targets.
Technology and Data Variations List	• Targets may be defined per employee, per department, or per store. • Data may be entered manually or uploaded via CSV file.
Frequency of Occurrence	• Medium – Typically done once every month.
Miscellaneous	• Historical targets should remain stored for performance comparison.

UC06

Use Case Section	Comment
Use Case Name*	Review Employee Performance
Scope	SmartKiryana.
Level	User Goal
Primary Actor*	Admin
Stakeholders and Interests	**Admin:** Wants insight into employee productivity. **Organization:** Needs performance monitoring.
Preconditions	• Employee sales data exists.
Success Guarantee (Post Conditions)	• Employee performance metrics are displayed.
Main Success Scenario*	**UC06: Main Success Scenario**

Actor Action	System Responsibility
1. Admin opens employee performance module.	
	2. System displays available employee list.
3. Admin selects an employee	3. System retrieves performance data.
4. Admin requests performance analysis.	4. System calculates performance metrics.
	5. System displays performance charts and statistics.

Extensions*	Employee data unavailable • System displays notification: No performance data

	available.
Special Requirements	• Performance metrics should include sales totals, target achievement %, and rankings. • Graphical representation should be supported.
Technology and Data Variations List	• Performance may be viewed per employee or time. • Data may be displayed in charts, tables, or dashboards.
Frequency of Occurrence	• Medium – Admin may review performance weekly or monthly.
Miscellaneous	• Future versions may include AI-based performance recommendations.

UC07

Use Case Section	Comment
Use Case Name*	View Sales Dashboard
Scope	SmartKiryana
Level	User Goal
Primary Actor*	Admin
Stakeholders and Interests	**Admin:** requires quick visualization of business performance.
Preconditions	• Admin is logged into the system.
Success Guarantee (Post Conditions)	• Dashboard with updated sales analytics is displayed.
Main Success Scenario*	**UC07: Main Success Scenario** Actor Action / System Responsibility 1. Admin selects the dashboard option. / / 2. System loads dashboard interface. 3. Admin requests dashboard analytics. / 3. System retrieves sales analytics data. / 4. System displays charts, graphs, etc.
Extensions*	Analytics data unavailable • System displays message indicating data retrieval failure.
Special Requirements	• Dashboard must refresh data automatically at regular intervals.

	• Charts should be interactive and responsive
Technology and Data Variations List	• Dashboard may include bar charts, pie charts, line graphs, and KPI indicators. • Data sources may include real-time sales transactions and stored analytics data.
Frequency of Occurrence	• Very High – Admin may access the dashboard several times daily.
Miscellaneous	• Users may customize dashboard widgets in future updates.

UC08

Use Case Section	Comment
Use Case Name*	Generate Monthly Reports
Scope	SmartKiryana
Level	User Goal
Primary Actor*	Admin
Stakeholders and Interests	**Admin:** requires summarized reports for evaluation and planning.
Preconditions	• Monthly sales data exists in the system.
Success Guarantee (Post Conditions)	• A monthly sales report is generated and displayed.
Main Success Scenario*	**UC08: Main Success Scenario**

Actor Action	System Responsibility
1. Admin opens the report generation module.	
	2. System displays report options.
3. Admin selects the month and report format.	3. System validates input parameters.
4. Admin requests report generation.	4. System processes sales data.
	5. System generates monthly report.
	6. System displays the generated report.

	7. Admin views or downloads report.	
Extensions*	Data missing for selected month • System displays message indicating insufficient data.	
Special Requirements	• Reports must support export formats such as PDF and Excel. • Reports should include sales summaries, charts, and performance indicators.	
Technology and Data Variations List	• Reports may be generated for different months, stores, or product categories. • Output formats may vary (PDF, Excel, or on-screen display).	
Frequency of Occurrence	• Medium – Typically generated once per month.	
Miscellaneous	• Reports may later be automatically emailed to management.	

UC09

Use Case Section	Comment
Use Case Name*	Process Sale
Scope	SmartKiryana
Level	User Goal
Primary Actor*	Employee
Stakeholders and Interests	• **Admin:** Wants correct revenue recording and inventory deduction. • **Employee:** Wants a fast, accurate billing process.
Preconditions	• Employee is logged into the system. • Products are registered in the system.
Success Guarantee (Post Conditions)	• A complete and accurate bill is generated. • The inventory is automatically updated to show that the items have been sold. • The transaction is recorded in the system's bill history.
Is Main Success Scenario*	**UC09: Main Success Scenario**

Actor Action	System Responsibility
1. Employee starts a new billing session.	
2. Employee scans barcode, enters serial number, or manually searches for the item. (Repeated for each item)	2. System retrieves item details and adds item to the bill with a running total
3. Employee reviews bill and adjusts quantities if needed.	3. System recalculates totals after each modification.
	4. System presents final total with applicable taxes.
5. Employee confirms the bill and proceeds to checkout	5. System finalizes the bill and saves the transaction.

Extensions*	• Item not found by barcode or serial number:
	  ○ System displays an error message. Employee switches to manual search.
	• Database not connected
	  ○ Reload the Inventory Module

Special Requirements	• System must respond to item addition within 2 seconds. • Billing must maintain 99.9% calculation accuracy.
Technology and Data Variations List	• Items may be added to the receipt using a barcode scanner, keyboard serial number entry, or mouse-manual search. • Receipts are printed physically or stored as electronic records.
Frequency of Occurrence	• Multiple times per hour during store operating hours.
Miscellaneous	• Could I include customer loyalty points for discount?

UC10

Use Case Section	Comment
Use Case Name*	Process Item Return
Scope	SmartKiryana
Level	User Goal
Primary Actor*	Employee
Stakeholders and Interests	**Admin:** Wants accurate inventory restoration and return records. **Employees**: Wants a straightforward process to handle customer returns.
Preconditions	• Employee is logged into the system. • The original transaction exists in the bill history.
Success Guarantee (Post Conditions)	• Returned items are restored to inventory. • The return is recorded in the system's transaction history. • Revenue records are adjusted accordingly.
Main Success Scenario*	**UC10: Main Success Scenario**

Actor Action	System Responsibility
1. Employee accesses the bill history and locates the original transaction.	
	2. System displays the original bill details including all purchased items.
3. Employee selects the item to be returned and initiates the return.	3. System validates the return request.
4. Employee confirms the return.	

	5. System processes the return, restores stock levels, records the return transaction, and updates revenue figures.

Extensions*	• Original transaction cannot be found:
	  ○ System notifies the Employee. Employee may attempt a manual search by date or amount.
	• Database or System Error
	  ○ Ask the Employee to try again

Special Requirements	• Return processing must update inventory in real time. • All return records must be auditable by Admin.
Technology and Data Variations List	• Return confirmation may be printed or recorded digitally.
Frequency of Occurrence	• Occasional. several times per day depending on customer activity.
Miscellaneous	• Can the admin approval be needed for high-value returns?

UC11

Use Case Section	Comment
Use Case Name*	Generate Customer Bill
Scope	SmartKiryana.
Level	User Goal
Primary Actor*	Employee
Stakeholders and Interests	• **Employee:** Wants to quickly generate an accurate bill for the customer. • **Admin:** Wants all generated bills stored accurately for reporting. • **Government Regulators:** Want tax-inclusive receipts for audit purposes
Preconditions	• A billing session is active with at least one item added.
Success Guarantee (Post Conditions)	• A formatted receipt is created that includes all items ordered, their quantities, prices, taxes, and total amount. • The bill is recorded in the bill history. • The stock is updated to reflect the sale.
Main Success Scenario*	**UC11: Main Success Scenario**

Actor Action	System Responsibility
1. Employee requests bill generation.	
	2. System calculates the subtotal, applies applicable taxes, and computes the final total.
	3. System generates a formatted receipt.
4. Employee selects the receipt delivery method (print or digital).	4. System outputs the receipt via the selected method.

	5. System saves the bill on the transaction history and deducts sold items from inventory.
Extensions*	• Printer is unavailable: ○ System notifies the Employee. Employee switches to digital receipt option.
Special Requirements	• Receipt generation must complete within 3 seconds. • Receipt must be formatted clearly and include all legally required fields.
Technology and Data Variations List	• Receipts may be printed via a thermal printer or stored as a digital record in the system.
Frequency of Occurrence	• Every completed billing session, multiple times per hour.
Miscellaneous	• Can the system email or SMS the digital receipts to customers?

UC12

Use Case Section	Comment
Use Case Name*	View Inventory Availability
Scope	SmartKiryana
Level	User Goal
Primary Actor*	Admin, Employee
Stakeholders and Interests	**Admin:** Wants accurate, real-time visibility of stock levels to make restocking decisions. **Employee:** Wants to quickly check item availability during customer inquiries.
Preconditions	• User is logged into the system with appropriate access rights.
Success Guarantee (Post Conditions)	• The actor views accurate, real-time stock levels for the requested products. • Any understock or overstock alerts are clearly displayed.
Main Success Scenario*	**UC12: Main Success Scenario**

Actor Action	System Responsibility
1. Actor navigates to the Inventory Management module.	
	2. System loads and displays the full inventory list.
	3. System highlights any items flagged as understocked or overstocked.

Extensions*	• Inventory data fails to load:
	  ○ System displays an error notification
	  ○ If the issue persist contact the IT department.

Special Requirements	• Inventory data must reflect real-time stock levels updated after every transaction. • Stock alerts must be visually distinct (e.g., color-coded) for quick identification.
Technology and Data Variations List	• Inventory may be searched by product name, category name, or barcode number.
Frequency of Occurrence	• Multiple times, as needed by user during customer interactions.
Miscellaneous	• Should the system support automated purchase order generation when understock is detected?

UC13

Use Case Section	Comment
Use Case Name*	Access Bill History
Scope	SmartKiryana
Level	User Goal
Primary Actor*	Admin, Employee
Stakeholders and Interests	• **Admin:** Wants access to all transactions for auditing, reporting, and analyzing. • **Employee:** Wants access to certain bills for processing returns or settling customer disputes. • **Government Regulators:** Wants access to auditable transaction records for verifying taxes.
Preconditions	• User is logged into the system. • At least one completed billing transaction exists in the system.
Success Guarantee (Post Conditions)	• The actor successfully retrieves and views the required transaction records. • All displayed bill details are accurate and complete.
Main Success Scenario*	**UC13: Main Success Scenario**

Actor Action	System Responsibility
1. User navigates to the Bill History module.	
	2. System loads and displays a list of completed bills.
3. Actor applies filters (e.g., by date range etc.) to narrow results	
	4. System filters and displays the matching bill records.
5. Actor selects a specific bill to view its full details.	

	6. System displays the complete bill including all items, quantities, prices, tax breakdown, and total.

Extensions*	• Bill history fails to load:
	  ○ System displays an error notification.
	  ○ User contacts IT Support.

Special Requirements	• Bill history must be retrievable within 3 seconds for any date range. • All records must be read-only to preserve audit integrity.
Technology and Data Variations List	• Admin may export bill history as a report (e.g. PDF etc).
Frequency of Occurrence	• Several times daily by Admin for reporting and by Employees when processing returns.
Miscellaneous	• Can I apply long-term archiving strategy for historical bills?

UC14

Use Case Section	Comment
Use Case Name*	Track Personal Performance
Scope	SmartKiryana
Level	User Goal
Primary Actor*	Employee
Stakeholders and Interests	**Employee:** Wants to monitor their sales performance and progress toward targets to improve productivity.
Preconditions	• The employee must be successfully logged into the system. • The employee's sales records must exist in the database.
Success Guarantee (Post Conditions)	• The employee is shown a summary report of their personal performance.
Main Success Scenario*	**UC14: Main Success Scenario**

Actor Action	System Responsibility
1. The employee logs into the system and navigates to the Personal Performance section.	
	2. The system retrieves the employee's sales records from the database and calculates relevant performance metrics.
	3. The system generates a performance summary report and displays it to the employee.

Extensions*	• No Sales Data Available

	◦ The system will prompt a message that there is no performance data available. • System Failure ◦ Ask the user to restart the system and try again. ◦ If the problem still exists, the user should contact the IT department.
Special Requirements	• The system should be able to generate the performance report in 3 seconds or less. • Only the employee should be able to access his/her performance report. • The performance report should be presented in an easily readable format.
Technology and Data Variations List	• The performance report may be displayed as a table, graphical chart, or downloadable report.
Frequency of Occurrence	• Occasionally – Used When the employee wants to see their performance only
Miscellaneous	• Should employees receive automatic notifications when they reach sales milestones? • Should historical performance reports (e.g., previous months) also be available?

UC15

Use Case Section	Comment
Use Case Name*	Manage Product Categories
Scope	SmartKiryana
Level	User Goal
Primary Actor*	Admin, Employee
Stakeholders and Interests	• **Admin:** Wants to manage and organize product categories in an efficient manner to ensure a well-structured inventory. • **Employee:** Wants to access well-organized product categories in order to easily identify the product during inventory or sales operations.
Preconditions	• The user must be logged into the system.
Success Guarantee (Post Conditions)	• The product category records are updated based on the operation performed. • The list of product categories is shown to the user. • A message is shown to the user indicating that the operation was successful.
Main Success Scenario*	**UC15: Main Success Scenario**

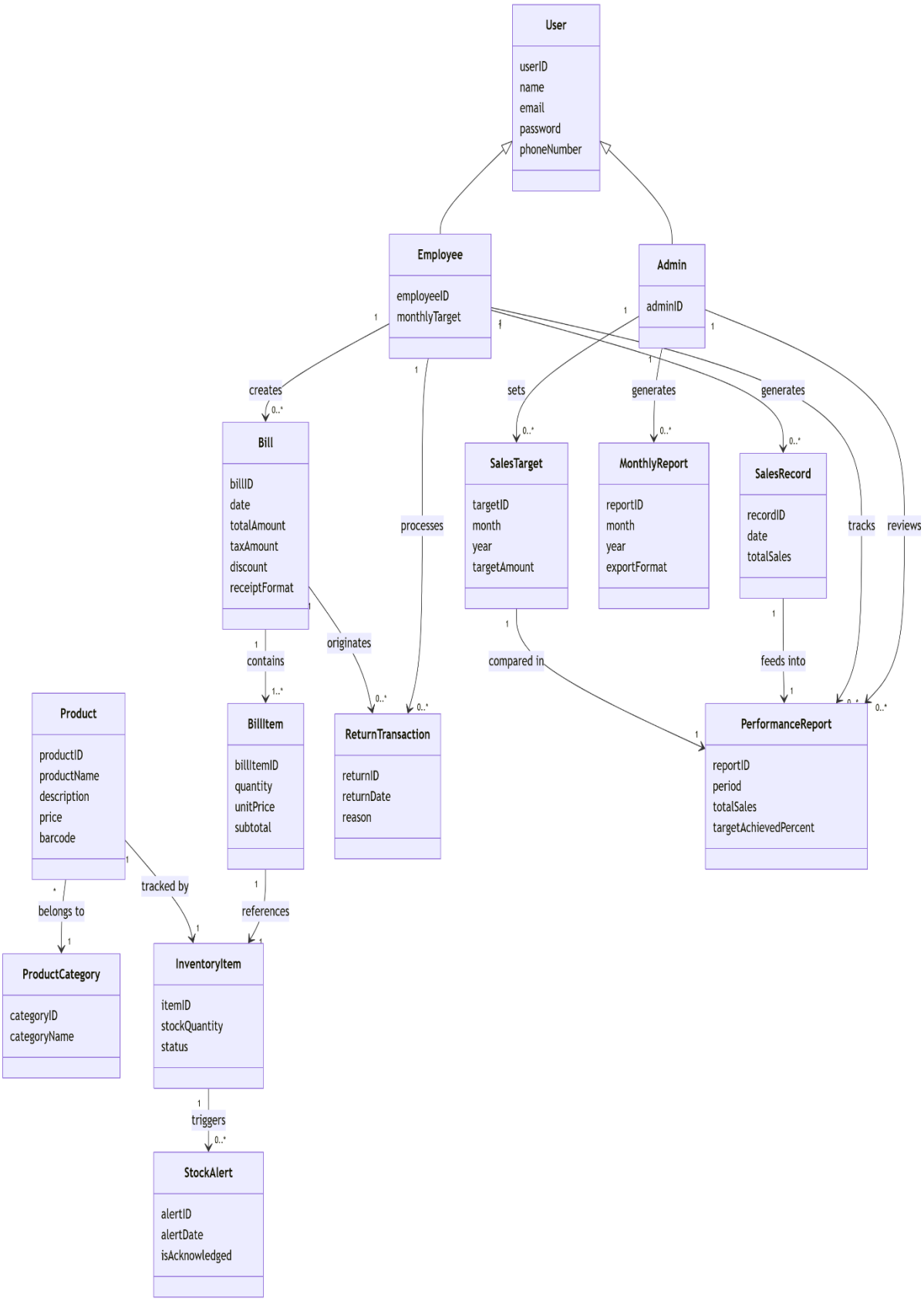
Actor Action	System Responsibility
1. The user logs into the system and chooses the Product Category Management option.	
	2. The system retrieves and displays the current list of product categories and available operations.
3. The user selects a category operation and	3. The system validates the requested Operation and data

	enters the required Data.	entered.
		4. The system updates the category records in the database.
		5. The system displays a success message and an updated list of product categories.

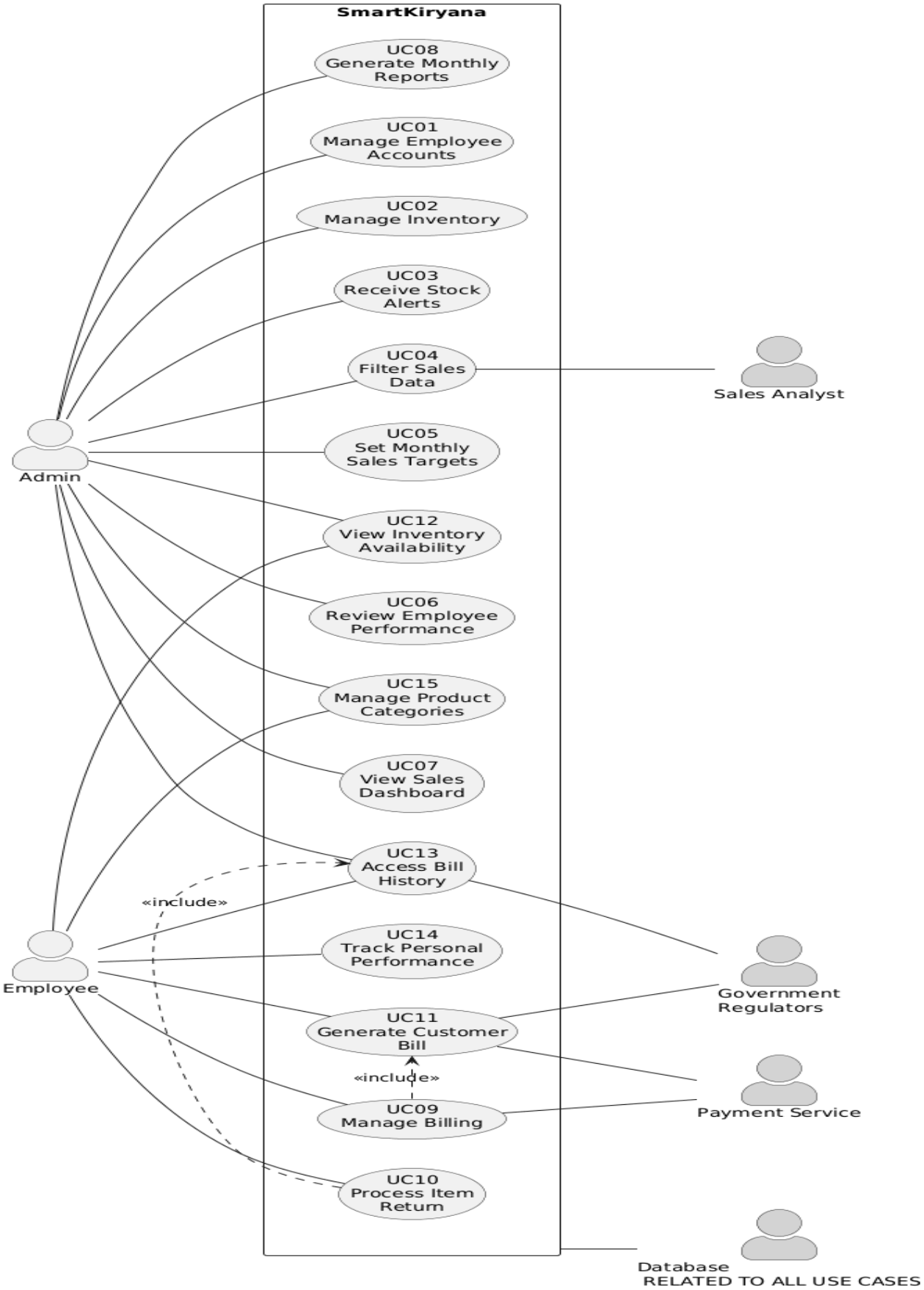
Extensions*	• Invalid Category Data
	  ○ The system generates an alert that the entered data is invalid
	• Duplicate Category Name
	  ○ The system generates an error that the category already exists.	

Special Requirements	• Only authorized users should be allowed to update category records. • The list of categories should be prominently displayed. • The categories should be kept consistent with the product inventory system.	
Technology and Data Variations List	• The Categories may be based on the same company (Barcode) or would be according to the nature of product(manual)	
Frequency of Occurrence	• Medium – Typically used once per week.	
Miscellaneous	• Should categories support icons or images for easier identification? • Should employees have restricted permissions compared to admins (e.g., employees cannot delete categories)?	

Domain Model:



Use Case Diagram:


SmartKiryana
UC08 Generate Monthly Reports
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UC15 Manage Product Categories
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UC13 Access Bill History
UC14 Track Personal Performance
UC11 Generate Customer Bill
UC09 Manage Billing
UC10 Process Item Return
«include»
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Admin
Employee
Sales Analyst
Government Regulators
Payment Service
Database
RELATED TO ALL USE CASES